

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on

the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted [online](#).
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: **Finland**
- Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: **Ministry of Education and Culture**

Website: <https://okm.fi/en/frontpage>

Email address*:

Phone number*:

Please describe the role/mandate of your organization:

The Ministry of Education and Culture (MOEC) oversees science policy and higher education policy in Finland.

- Officially designated contact person(s) who completed this survey:

Full name*: Hannu Vainonen

Position*: Senior Ministerial Adviser

Email address*:

- Full Name(s) of designated official(s) certifying the report:

Hannu Vainonen, Sami Niinimäki, Juha Haataja

- Brief description of the consultation process established for the preparation of the report:

MOEC consulted the Research Council of Finland, Business Finland and The Federation of Finnish Learned Societies.

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Federation of Finnish Learned Societies (TSV)

Website*: <https://www.tsv.fi/en>

Email address*

Sector: ☐ Public ☐ Private ☒ Civil Society ☐ Other

Name of the organization*: Research Council of Finland

Website*: <https://www.aka.fi/en>

Email address*

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

Name of the organization*: Business Finland

Website*: <https://www.businessfinland.fi/en/for-finnish-customers/home>

Email address*

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

If **yes**, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The Ministry of Education and Culture.

The Research Council of Finland.

Business Finland.

Secretariat for the National Open Science and Research Coordination within the Federation of Finnish Learned Societies, [link to website](#).

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

If **yes**, which language(s)?

YES

Unofficial translation in Finnish is available on the website of the Secretariat for the National Open Science and Research Coordination. [Link](#)

Swedish is the other official language in Finland. The Swedish translation of the UNESCO recommendation is available on the website of Svenska Uneskorådet. [Link](#)

- 1.3 Have there been awareness raising activities on open science, including all its key elements¹, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

If **yes**, please provide details on the activities organized or foreseen and links as relevant.

YES

In 2018, the Ministry of Education and Culture awarded funding for TSV to coordinate and promote open science (Ministry of Education and Culture, [link](#), in Finnish). Since 2018, the activities of National Open Science and Research Coordination are supported by the Secretariat for the National Open Science and Research Coordination within TSV (Source: TSV, [link](#), in Finnish). The Secretariat is financed by the Ministry of Education and Culture ([link](#)).

National Open Science and Research Coordination has actively promoted open science throughout its operating time from 2018 onwards. A summary of the activities is available in this update on the open science website, [link](#).

The Research Council of Finland (RCF) has long-established practices ~~is~~ committed to promoting and supporting open and responsible science. The science policy priorities and practical measures have depended on the timely practices of the scientific community. The RCF policies are aligned with international development, i.e. Plan S., CoARA - Coalition for

Advancing Research Assessment and Science Europe. Multiple measures to raise the awareness of the applicants and grantees on these have been taken.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

YES

Research Council of Finland

Research Council of Finland (RCF) requires that RCF-funded projects commit to ensuring immediate open access to their peer-reviewed articles in accordance with [Plan S principles](#) and [Finland's national policy for open access to scholarly publications](#). This requirement applies to projects funded from RCF calls opened after 1 January 2021. The Research Council also urges researchers to publish their conference articles and monographs with open access. (Source : Research Council of Finland, [link](#).)

The Research Council requires that principal investigators of projects funded by the Research Council be responsible for the responsible management and opening of research data. Research data must be made freely available as soon as possible after the research results have been published. Sites of research must therefore provide researchers with the necessary guidance and ensure that they have access to suitable storage infrastructure. Research data shall be managed and made available following [the FAIR principles](#) (F = findable, A = accessible, I = interoperable and R = reusable). (Source : Research Council of Finland, [link](#).)

Business Finland

Business Finland expects that the results of its public research funding projects are public after the completion of the project. The beneficiary must publicize the results actively, broadly and without discrimination. (Source: Business Finland, [link](#))

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

- does it include any of the following open science elements:
 - open access to scientific knowledge
 - open science infrastructure
 - open engagement of societal actors
 - open dialogue with other knowledge systems

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

In line with the [Act on Research and Development Funding in 2024–2030 \(1092/2022\)](#), the Finnish Government shall adopt a plan for the use of central government research and development funding once every electoral term (current term is 2023-2027). [The multiannual plan for the use of government research and development funding](#) published in 2024 presents the current state of the research, development and innovation system, the objectives of the system and the main policies for the use of R&D funding. Decisions on the actual allocation of funding are made as part of the normal budgetary process.

On 9 October 2023, the Ministry of Education and Culture and the Ministry of Economic Affairs and Employment appointed a working group to draft a proposal for the plan. The working group was responsible for drawing up the proposal for the plan and its monitoring mechanisms and for holding consultations with stakeholders. The wide-ranging group of stakeholders was tasked with supporting and monitoring the preparations, producing expert information and presenting views on the policies in the plan. The preparations were led by the [Research and Innovation Council](#) and involved extensive consultation with the stakeholders.

The main policy priorities identified in the plan involve increasing cooperation between R&D actors, promoting ambitious R&D activities in businesses, strengthening competence, improving research and technology infrastructure, deepening EU and other international cooperation and making strategic choices. The plan also describes the expected impacts and presents indicators to support monitoring.

The plan does not include any of the mentioned open science elements.

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

YES

In Finland, the [Declaration of Open Science \(2020\) and Research and its supporting policies](#) have been developed by the Finnish research community (see Q 2.4).

The Declaration is supported by four policies, one in each of the areas of open science, as well as one policy component common for two areas, number 5 in the below list:

1. Policy for open scholarship (2022), [link](#)
2. Open access to scholarly publications. National Policy and executive plan by the research community in Finland for 2020–2025 (2019), [link](#)
3. Open research data and methods. National policy and executive plan by the higher education and research community for 2021–2025: Policy component 1 (Open access to research data) (2021) and 2 (Open access to research methods and infrastructures) (2023), [link](#)
4. Open education and educational resources. National policy and executive plan by the higher education and research community for 2021–2025. Policy components 1 (Open access to educational resources) (2020) and 2 (Open educational practices) (2022), [link](#)
5. Open education and educational resources, open access to scholarly publications. National policy and executive plan by the higher education and research community for 2021–2025: Policy component for open access to theses (2024), [link](#).

There is also some national legislation supporting the principles of open sciences:

[The Finnish Act on the Openness of Government Activities \(621/1999\)](#), ensures the transparency of public administration (including public expert and research institutes) by granting individuals the right to access/request official documents held by authorities. It establishes the principle of openness while also specifying exceptions to protect privacy, national security, and other sensitive information.

The Directive on Open Data and Public Sector Information (Directive (EU) 2019/1024) was implemented through national legislation: [Act on the Reuse of Publicly Funded Research Data \(713/2021\)](#). The act applies to research data that is publicly funded and made publicly available through a data repository by researchers, research organisations, or organisations funding research.

[Data Protection Act \(Tietosuoja laki 1050/2018\)](#) implements the EU's General Data Protection Regulation (GDPR) in Finland. It governs how personal data is collected, processed, and shared in research.

[Statistics Act \(Tilastolaki 280/2004\)](#) governs the use and sharing of statistical data produced by public institutions in Finland.

[Copyright Act \(404/1961, amendments up to 1216/2023\)](#) has provisions on text and data mining. Research organisations and cultural heritage institutions with lawful access to a work may reproduce it in order to carry out text and data mining in scientific research and retain the copies for the purpose of scientific research, including for the subsequent verification of research results, provided that the copies are only available to those with lawful access to them. These provisions shall not be derogated from in contracts and agreements, nor shall its application be prevented by technical means.

2.3 In your country, are there specific policy instruments^v that aim to promote open science?

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

YES

University funding

Title of the instrument: Finnish universities' core funding

Objective: Steering of higher education institutions

The authority in charge: Ministry of Education and Culture

Allocated budget: There is no predefined budget for funding based on open publications in the universities.

Elements of open science included: open access to peer-reviewed research articles.

In the funding period 2021–2024, 14 % of the Finnish universities' core funding is based on scientific publications. The coefficient for open publications is 1.2. (Source: Ministry of Education and Culture, [link](#).)

In the funding period 2025–2028, 14 % of the Finnish universities' core funding is based on scientific publications. The coefficient for open publications is 1.2. (Source: Ministry of Education and Culture, [link to memo with English abstract](#), [link to the act in Finnish](#).)

Funding of the universities of applied sciences

Title of the instrument: Finnish Universities of Applied Sciences Core Funding

Objective: Steering of higher education institutions

The authority in charge: Ministry of Education and Culture

Allocated budget: There is no predefined budget for funding based on open research outputs in the universities of applied sciences.

Elements of open science included: open access to research outputs.

In the funding period 2021–2024, 2 % of core funding for the Finnish universities of applied sciences is based on publications, public artistic and design activities, audiovisual material and

ICT software. The coefficient for open publications is 1.2. (Source: Ministry of Education and Culture, [link](#).)

In the funding period 2025–2028, 2 % of core funding for the Finnish universities of applied sciences is based on publications, public artistic and design activities, audiovisual material and ICT software. The coefficient for open publications is 1.2. (Source: Ministry of Education and Culture, [link to memo with English abstract](#), [link to the act in Finnish](#).)

Research Council of Finland

Title of the instrument: All funding programmes. One example is the FIRI 2021 call for research infrastructures on the 2021–2024 roadmap and Finland's memberships in international research infrastructures (FIRI2021 Roadmap and international memberships)

Objective: To subsidize research infrastructures

The authority in charge: Research Council of Finland

Allocated budget: 30 M€

Elements of open science included: In the evaluation of the applications the following criterion is taken into consideration: In its activities, the research infrastructure must take into account research ethics, equality and non-discrimination, the principles of open science and the sustainable development goals. The project must follow these same principles. ([link](#))

Other

Title of the instrument: Grants for publishing activities of the learned societies

Objective: To subsidize national publishing activities of the learned societies

The authority in charge: The Ministry of Education and Culture delegates funding to be used by the Federation of Finnish Learned Societies.

Allocated budget: In 2025 1,22 M€.

Elements of open science included: open access to research articles. The grants delegated by the Federation of Finnish Learned Societies for publishing purposes need to be compliant with the funding terms of the Research Council of Finland. ([link](#))

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\)](#) [17.a](#), [17.b](#), [17.c](#), [17.f](#), [17.h](#))

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

YES

In Finland, **the Declaration of Open Science and Research** and its supporting policies have been developed by the Finnish research community. By signing the declaration for open science and research 2020–2025, organisations and other members of the research community commit to the vision and mission, including supporting and encouraging the everyday work of their organisation's members to attain the objectives and goals of the open science and research policies. (Source: Open Science and Research Coordination, [link](#); National Open Science and Research Coordination, [link](#).)

In implementing the declaration and its supporting policies, the research community has executive responsibility while the TSV has coordinating responsibility. (Source: Open Science and Research Coordination, [link](#).)

Policies of open science and research in Finland outline in detail the strategic principles, objectives and action plans necessary to achieve the objectives set out in the Declaration for Open Science and Research. (Source: Secretariat for the National Open Science and Research Coordination, [link](#).)

The policies described in 2.2 contain the objectives and actions as well as the roadmap for the implementation of open science. There are objectives and actions for different types of actors, including research performing organizations, research funding organizations and other types of organizations involved, e.g. in supporting OS on the national level like the Secretariat for the National Open Science and Research Coordination.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

YES /UNDER DEVELOPMENT/ NO

If **yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#), included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

YES

Data from the Finnish [survey of the monitoring of open science and research](#). Of the Finnish universities, universities of applied sciences and research institutes, 44 have one or more openly published documents that guide open science and research throughout the organization. Additionally, two organizations have a document that has not been openly published. Examples: [University of Helsinki](#), [Turku UAS](#) and [Natural Resources Institute of Finland](#).

Policy documents in Finnish RPOs in the national OS monitoring 2024,
Total number of respondents = 47

Policy	Openly published	Exists but is not openly published	In preparation and published within 12 months	No
Culture for open scholarship	44	2	n/a	1
Citizen science	12	1	14	20
Collaboration between research organizations and companies	22	9	9	7
Open access publishing	43	4	n/a	0
Openness of research data	43	3	n/a	1
Openness of infrastructures	24	4	n/a	19

Numbers in columns refer to the number of responding organisations.
Only new policies have the option “in preparation” as an option in the survey.

Scientific unions and associations and learned societies are not included in the national OS monitoring. Promoting open availability of scientific knowledge and publications is one of the focus points in the strategy of the Federation of Finnish Learned Societies (TSV) ([link](#), in Finnish only). TSV is a national co-operative body for learned societies in Finland. The Federation has a membership of 295 societies. ([link](#))

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level?
(ref.: [\(ii\) 17.i](#), [\(iii\) 18.i](#), [\(iv\) 19.c](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

UNDER DEVELOPMENT

[There is a working group established by the National Steering Group on Open Science and Research with the objective of designing a funding model for open science and creating a supporting action plan for all areas of openness.](#) The working group met for the first time on May 30, 2023, and has a two-year term of office. [The first version of the road map](#) has been subjected to a public round of comments from 28 October until 5 December 2024. The output of this working group will be a recommendation.

Also see replies to Q 1.4 and Q 2.3.

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

See reply to Q 1.4 and 2.3.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

There is a national [Open science recommendation and checklist for research, development and innovation activities in collaboration between research organisations and companies](#), published in 2022.

The [2021 Updated National Roadmap for Research, Development and Innovation](#) emphasized the importance of public-private partnerships, particularly in response to challenges like the COVID-19 pandemic. The ministries pledged to promote data utilisation through open science, regulation, agreements and industry selfregulation, drawing on European and international cooperation. The period for this roadmap ended in 2023.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.i](#), [23.c](#))

If yes or under development, please provide details and links as relevant.

YES

In Finland, national monitoring of the current framework for open science and research has been performed in 2022 and 2024. The results of these monitoring implementations are available in Research.fi portal. (Source: Research.fi, [link](#)). There is a summary of the results available on the Secretariat for the National Open Science and Research Coordination website ([link to 2024 results](#) and [2022 results](#)).

National open science and research monitoring is based on the Monitoring model for open science and research (Source: Open Science and Research Coordination, [link](#)).

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

In 2023, the share of research and development expenditure was 3,09% of GDP.

Date of publication 24 Oct 2024, source Statistics Finland, [link](#).

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

In 2023, a total of 94 per cent of Finns aged 16 to 89 used the Internet. (Source: Statistics Finland, release date 12/12/2023, [link](#))

At the end of September 2023, a fast fixed connection with a maximum download speed of at least 100 megabytes was available to 78% of Finnish households. The availability of a 1 GB network and covered 71% of households. (Source: Finnish Transport and Communications Agency Traficom, [link](#))

High-speed 4G or 5G mobile networks covered 96% of Finnish households at the end of June 2024. (Source: Finnish Transport and Communications Agency Traficom, [link](#))

Note: ITU provides statistics on individuals Internet usage and household Internet access, [link](#). The Finnish statistics follow this same structure.

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

YES

Funet (Finnish University and Research Network) is an information network for Finnish higher education institutions, research and education. Funet offers fast and reliable data connections and cost-effective Funet services to universities, universities of applied sciences, public research institutes and other organisations that meet the Funet membership criteria. Funet is operated by CSC – IT Center for Science Ltd.

The Funet network enables learners and scientists to connect to national and international research networks and internet via the Nordic [NORDUnet](#) and European [GÉANT](#). (Source: CSC, [link](#).)

All Finnish universities and universities of applied sciences are members of Funet. In addition, Funet has 39 other members, including e.g. state research institutes and other government agencies, private foundations and university hospitals. (Source: [info.funet.fi](#), in Finnish only)

3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The Finnish national roadmap for research infrastructures (FIRI) 2021-2024 includes 29 research infrastructures (RI) that are funded by the Research Council of Finland. (Source: Research Council of Finland, [link](#).)

These research infrastructures are not open science specific. Yet the Research Council of Finland expects that the services and data produced with the RI are openly available. The RI must be open and accessible for all researchers. (Source: Research Council of Finland, [link](#), in Finnish).

Example of a federated IT infrastructure that supports open science and is included in the FIRI roadmap are CSC's Research Infrastructure Services. They include

- Platform for data management, data analytics and scientific computing
- National research and education network Funet, see also reply to Q 3.3
- Supercomputing, see also reply to Q 3.5
- Object storage environment to store, share or move research data
- FAIRdata service portfolio, which has been especially designed for Open Science, includes:
 - Fairdata IDA – research data storage
 - Fairdata Qvain – research dataset description tool
 - Fairdata Etsin – research data finder
 - Fairdata AVAA – open data publishing platform
 - Fairdata Metax – metadata warehouse
 - Digital Preservation Service for Research Data

The Fairdata services are developed in accordance with the FAIR principles.

(Source: Research Council of Finland, [link](#).)

CSC is not a commercial actor and has other sources of financing besides RCF. CSC is a company entrusted with special state assignment and owned by the state of Finland (70% of the shares) and Finnish higher education institutions (30% of the shares). [link](#)

Companies entrusted with special state assignments are companies owned either partly or completely by the state. The company has an assignment defined, regulated and overseen by the state.

CSC's services fit into the category federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.

For this survey, the [Research.fi](#) service is used as an example of the CSC services.

inclusivity: The [Research.fi](#) service contains information from e.g. the Finnish research system, publications by Finnish organizations, projects funded by public and private research funders, as well as statistical information on the development of research staff and funding figures and the impact of scientific publications. ([link](#))

accessibility: The accessibility requirements for the online service are based on the Act on the Provision of Digital Services (306/2019). Research.fi meets the level AA critical accessibility requirements defined in WCAG 2.1. Only the statistical contents of the portal's "Figures on science and research" section can be problematic for users of screen readers. ([link](#))

interoperability: Data is automatically collected from organisation to the National Research Information Hub. Research.fi is the external service of the Research Information Hub that provides easy access to the information. ([link](#))

At a later stage, the goal is to make it possible to also download information in the service to a file or retrieve it through a machine-readable API interface. ([link](#))

community governance: Finnish higher education institutions own 30% of the shares in CSC, see above.

FAIR (Findable, Accessible, Interoperable, and Reusable): The service uses national [Fairdata services](#) as a source of information for research data. Fairdata services contain information on

Finnish research data, including metadata saved by researchers themselves as well as transferred from research organisations. ([link](#))

CARE (Collective Benefit, Authority to Control, Responsibility and Ethics): Research.fi portal has publications in the indigenous Same language. These publications can be found through site search. ([link](#)) The portal provides links to the publications, it does not act as a repository. ([link](#))

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\)](#) [18.e](#), [18.f](#))

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open scienceⁱ, stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES

The European High-Performance Computing Joint Undertaking (EuroHPC JU) is pooling European resources to develop top-of-the-range exascale supercomputers for processing big data, based on competitive European technology.

One of the pan-European pre-exascale supercomputers, LUMI, is located in CSC's data center in Kajaani, Finland.

The supercomputer is hosted by the LUMI consortium. The LUMI (Large Unified Modern Infrastructure) consortium countries are Finland, Belgium, the Czech Republic, Denmark, Estonia, Iceland, the Netherlands, Norway, Poland, Sweden, and Switzerland.

Name of the infrastructure: Lumi HPC

LUMI is an HPE Cray EX supercomputer with a computing power of 380 petaflops, or 380 quadrillion operations per second, which is equivalent to the combined performance of 1.5 million laptop computers. At the start of its operation, LUMI was the third fastest supercomputer in the world. ([link](#))

accessibility to all the relevant users in the region or internationally: Lumi HPC can be used by researchers outside of Europe. By partnering with European research groups and/or companies, the EuroHPC resources are available to non-European research projects as well. (Note that the Principle Investigator for any projects that apply for time on LUMI needs to be based in the EU or an associated country.) ([link](#))

date of establishment: LUMI was officially accepted on 31 January 2023. ([link](#))

language: English

target groups and beneficiaries, governance/community agreements and scope:

Half of the LUMI resources belong to the EuroHPC Joint Undertaking, and the other half of the resources belong to the participating countries i.e. the LUMI consortium countries.

Each consortium country has a share of the resources based on the country's contribution to the LUMI funding. The shares for each of the countries are allocated according to local considerations and policies – so LUMI is seen and handled as an extension to national resources.

The LUMI shares belonging to the EuroHPC JU are allocated by a peer-review process (comparable to that used for PRACE Tier-0 access). In addition, up to 20% of the EuroHPC resources will be available to industry and SMEs.

A researcher affiliated with one of the LUMI consortium countries or a company which has its headquarters in the LUMI consortium countries is able to apply for LUMI resources via the LUMI consortium country's national share and, in addition, via EuroHPC's technical and scientific peer-review process. ([link](#))

LUMI HPC is not a dedicated open science resource. Yet it enables leading edge research and open science, e.g. Lumi-GPT-V: LumiGPT-Vision Multimodal Foundation Models will develop a family of massively multilingual and multimodal foundation models that can process both text and images and generate text in multiple languages as the output. The resulting models will enable a broad range of advances in multimodal AI across dozens of languages. The project will contribute toward transparent and trustworthy European AI by making all of its tools and models as well as their detailed training process openly available. ([link](#))

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

NO

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

Formal

Tampere University is piloting a training program for Data Stewards in 2025 ([link](#)). Training is arranged by the Unit Tree – Continuing Education. Its extent is 15 European Credit Transfer and Accumulation System (ECTS) credits.

This training is new and unique in its kind as Finland has lacked training aimed at the qualification of experts in the field of data management. The training has been designed and planned with extensive national cooperation. National data management operators and experts in the field have also been involved in the planning phase. Open science and FAIR data are included in this training.

The primary target group for this training is doctoral researchers. The training is also suitable for those who have already completed their thesis and/or who have already done some expert work in data management.

This training is funded by the Ministry of Education and Culture.

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

Already in the early stages of open science development, educational material for policy makers was developed, e.g. a handbook on open science and research in 2014 ([link, in Finnish only](#)).

More recent initiatives in this area with the Open Science Coordination are the working group Interaction with Decision-makers & Open Administration within the Culture of Open Science Expert Panel ([link](#)) and the recommendation The criteria for a good knowledge base (2024) ([link](#)). The recommendation is aimed at the drafters of legislation and comprises best practices for building a solid knowledge base on the basis of principles of responsible research. The key purpose of this recommendation is to supplement the guidelines for drafting government proposals.

Open Science Coordination also had a working group for Co-operation in national information policy under the expert panel of research data. The period of this working group ended in 2022. ([link, in Finnish only](#))

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

If yes, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

If this question refers to a national framework, it is not applicable to Finland as universities and universities of applied sciences have autonomy ([link to Universities Act](#), [link to Universities of Applied Sciences Act](#)).

However, based on replies to the question on a policy on skills/training for Open Science in the national OS monitoring 2024, 66 % of Finnish universities and universities of applied sciences had such a policy. This question is a part of the EOSC National Contributions to EOSC questionnaire and does not yield points in the national OS monitoring. It is included as data collection for the EOSC survey.

Replies to the question on the number of educational curricula with an Open Science dimension in 2022 in the national OS monitoring 2024 have variation. Some universities and universities of applied sciences replied only the number of curricula with OS dimension, others provided also the total number of curricula, and some replied simply “all curricula as OS is integrated to the courses supporting thesis work”. 9 of 13 universities and 11 of 23 universities of applied sciences provided either a numeric or textual response indicating their curricula include open science.

The file containing the full dataset of the national OS monitoring 2024 is not yet openly accessible. It will be made fully openly accessible in early 2025. Therefore, a link can be provided only in early 2025.

- 4.4 Are open educational resources^{*} as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES

In the national OS monitoring 2024,

- 89 % of the responding HEIs had a guiding document or documents on open educational materials.
- 89 % of the respondents recommend that open educational resources are published on an open platform
- 53 % of organisations have guidelines for learners on sharing the products of their learning openly.
- 47 % of organisations have a support person for learners on sharing the products of their learning openly.

Source: [Monitoring of open science 2024, results](#)

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES / NO

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

In 2018 Committee for Public Information in Finland released *Bold communication, responsible influence. Science communication recommendations*. ([link](#)) The recommendations are intended for everyone who works with science and those who communicate on science: science belongs to everyone. 4th recommendation addresses open science directly: “Support for science communication: High-quality, multifaceted science communication and science journalism build an open science culture and trust in science and research. Communication reaches different audiences.” The explanatory texts states that science communication is part of a culture of open science.

Finland has a [national CoARA chapter](#) (Coalition for Advancing Research Assessment) and 19 organizations with a CoARA action plan. One of the [core CoARA commitments](#) is to recognise the diversity of contributions to, and careers in, research in accordance with the needs and nature of the research. This commitment will broaden recognition of the diverse practices, activities and careers in research, considering the specific nature of research disciplines and other research endeavours. This includes science communication framed in the question.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\)](#) [20.b](#), [20c](#), [20j](#))

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

In Finland, promoting responsible research is an ongoing endeavour. The national recommendation “*Good practice in researcher evaluation. Recommendation for the responsible evaluation of a researcher in Finland*” ([link](#)) was approved in 2020. This recommendation has been drafted from the point of view of an individual researcher evaluation. The same principles should be followed when evaluating research organizations, research units, and research in a broader context. The recommendation is recommended to use in conjunction with The Finnish Advisory Board on Research Integrity’s (TENK) template for a researcher’s curriculum vitae.

In the section on the evaluation of research, the open science is one of the elements:

Open access to research Researchers' activities to promote open access to research outputs will be considered as part of the evaluation. Promoting open access is seen as part of the realisation of the fundamental values of research, societal impact, and the promotion of research. ([ibid.](#))

In the national open science monitoring 2024, 58 % of the responding RPOs had a policy document promoting that open science can be taken into consideration in assessments related to recruitment or career development or in other professional assessments, and 26 % plan to develop a policy in the next 12 months ([link to survey](#)).

In the same survey, in 64 % of the responding RPOs "*Recommendation for the responsible evaluation of a researcher in Finland*" has been taken into consideration and this has been reported as part of promoting responsible assessment within the organisation, and 23 % of the respondents intend doing so ([link to survey](#)).

Finnish Higher Education Institutes (HEI) and state research institutes are actively adopting the CoARA recommendation. The CoARA National Chapter Finland has 43 members, including 13 of 14 universities, 21 of 24 universities of applied sciences, 5 research institutes and the Research Council of Finland ([link](#)).

The FIN-CAM tool for researcher assessment is under development. Promoting open science is listed as one example in skills and capabilities. As examples in qualitative evaluation, the effectiveness and use of open publications, preprints, machine-readable data, open notebooks, methods or educational materials are mentioned. In quantitative evaluation, the number of open research outputs and funding received for open science and projects completed are listed as examples. (Link to the FIN-CAM model is forthcoming in January 2025 when public consultation is opened.)

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\)](#) [20.a](#), [20.c](#), [20.d](#))

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The Finnish National Board on Research Integrity TENK, Universities Finland UNIFI, Rectors' Conference of Finnish Universities of Applied Sciences Arene and the Academy of Finland prepared the first template for the researcher's curriculum vitae (CV) in accordance with the responsible conduct of research for Finnish research organisations in 2012. This curriculum vitae template was updated in 2020. The aim of the template is to provide guidelines for the writer of a CV so that the individual's merits are presented as comprehensively, truthfully and comparably as possible. TENK recommends that organisations adopt the template in its original format. However, specific instructions on how to use the template must be provided in each situation. ([link](#))

Section 14 in the TENK template addresses scientific and societal impact. It includes, e.g. promoting open science and research, for example the production and responsible distribution of research material and datasets, as well as key positions of trust, expert positions and

assignments and merits in research communication and appearing as an expert in the media. ([link](#))

In the national OS monitoring, one of the indicators assesses the use of the TENK CV template. The organisations must have enabled the alternative only if its use in evaluation is reflected in the organisation's recruitment, HR and / or evaluation guidelines and the person being evaluated has the opportunity to enter the necessary information into one of the platforms used in the evaluation. 51 % of all respondents have enabled providing information on other merits besides research outputs. By organisation type, there is much variation as the response rate was 100 % for the universities and 45 % for the research institutes and 26 % for the universities of applied sciences. ([link](#))

Steering-group for Responsible Researcher Assessment & Working Group is currently developing Career Assessment Matrix (FIN-CAM) of qualitative and quantitative criteria for researcher evaluation, which takes into account the diversity of research activities, the merits of open science ([link](#)) and differences in career stages, career paths and disciplines.

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The Finnish Open Science Coordination annually (since 2019) [presents several awards to recognize significant contributions to open science](#). The coordination of these awards is managed by the Federation of Finnish Learned Societies, with funding from the Ministry of Education and Culture. From 2015 to 2017, the Ministry also distributed annual awards for activities promoting research openness as part of the Ministry led Open Science and Research initiative.

Also see Q 5.1 on the Finnish development on responsible research assessment. The new assessment models will aim to recognize and reward open science practices.

- 5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20](#))

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

YES

Publication Forum

The steering group of the Publication Forum decided on 9.12.2024 to downgrade 271 journals from MDPI and Frontiers to level 0 from the beginning of 2025. The decision is based on the steering group's policy from 30.9.2024 to downgrade grey area journals from MDPI and Frontiers to level 0 from the beginning of 2025, excluding journals the panels suggested to keep at level 1 at their autumn meetings. Based on the panels' suggestions, 16 MDPI and 22 Frontiers journals will remain at level 1. ([link](#)) Grey area journals make use of the APC (Article Processing Charge) operating model and aim to increase the number of publications with the minimum time spend for editorial work and quality assessment. ([link](#))

Publication Forum expert panels decided to downgrade 60 journals to level 0 from the beginning of 2025 based on the research community's feedback on the quality assessment practices of scientific journals. In the spring of 2024, the Publication Forum collected research community's experiences on defects in quality assessment practices of scientific journals. When reviewing the feedback, the panels discussed the publisher's role in evaluations. Especially, the panels discussed commercial open access publishers, whose business model is based on collecting as many author fees (so called APC fees) as possible. ([link](#))

Publication Forum (in Finnish often referred to as JUFO) is a rating and classification system to support the quality assessment of research output. To account for the different publication cultures characteristic of various disciplines, the classification includes academic journals, book series, conferences as well as book publishers. The four-level classification rates the major foreign and domestic publication channels of all disciplines as follows:

1 = basic level

2 = leading level

3 = highest level

0 = publication channels that don't meet the criteria for level 1. ([link](#))

The evaluation is performed by 23 discipline-specific Expert Panels composed of some 300 distinguished Finnish or Finland-based scholars.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES, at both national and institutional levels.

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects

In Finland, citizen science is included in the policy for open scholarship ([link](#)). The policy is supported by the Recommendation for citizen science ([link](#)).

According to national open science monitoring 2024, 57 % of the responding research organisations had a policy on citizen science or a policy was in preparation ([link](#)). Also a number of services and training was available in citizen science ([link](#)). Legal services were offered equally with other projects in 62 % of the organisations, support and advice for general communication in citizen science was available in 57 % of the organisations and for the planning and implementing of citizen science in 43 % of the organisations. For personnel and researchers, participating in citizen science training as a part of other personnel training was possible in 40 % of the organisations.

- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.

The national roadmap for RDI (2020–2023), [link](#), had an action point that ministries develop the measurement of the effects of value creation related to the data utilisation and promote the opening and use of the data resources of public sector authorities in RDI activities. Also, the opening of public information resources was taken up as a way to create conditions for RDI activities and demand for innovations.

In Finland, there is a national network, Innokylä, for innovations in the healthcare and social services area ([link](#), in Finnish only). This innovation network is open for all actors in the healthcare and social services in public sector, associations, educational institutes and companies. The network has over 700 members. Its database has 417 entities and over 4,000 operating models. All outputs are openly available and reusable.

The network is coordinated by [the Finnish Institute for Health and Welfare \(THL\)](#) and [SOSTE Finnish Federation for Social Affairs and Health](#).

- Others (with a box to add text)

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

Finland is participating in the European Open Science Cloud (EOSC), the Research Data Alliance (RDA), International Science Council Committee on Data (CODATA) communities, and European research infrastructure work. ESFRI, the European Strategy Forum on Research Infrastructures, is a strategic instrument to develop the scientific integration of Europe and to strengthen its international outreach. The competitive and open access to high quality Research Infrastructures supports and benchmarks the quality of the activities of European scientists, and attracts the best researchers from around the world.

Finland is participating in European Open Science Cloud Collaboration, for example by connecting the Fairdata services to the EOSC and hosting a national EOSC node (in preparation).

Finland is participating in the works to set up European Digital Infrastructure Consortia (EDICs) for the Genome EDIC, the Mobility and Logistics Data EDIC and Connected Public Administration EDIC (IMPACTS-EDIC). The country is finalising membership negotiations for the Local Digital Twins (LDT –CitiVERSE) EDIC and has decided to seek an observer status in the Alliance for Language Technologies one (ALT-EDIC)²(both already set up). The country also participates in the EuroHPC Joint Undertaking, hosting LUMI, one of the supercomputers procured by the Joint Undertaking.

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b](#), [22.e](#), [22.f](#))

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

Declaration for Open Science and Research (Finland) 2020–2025 and its supporting policies

The vision of the Declaration for Open Science and Research (Finland) 2020–2025, [link](#), states: “the Finnish research community is an international forerunner in open science and research”.

In the supporting policies, the policy for Open Scholarship Cooperation, [link](#), says in the actions of the first objective: “The organisation utilises national and international cooperation in the development of open science whenever possible and appropriate.”

The policy Open Education and Educational Resources, [link](#), states in its Objective 3: “By 2025 Finland is known as a pioneer and promoter of open educational resources in higher education.” Participating in international cooperation projects and disseminating information on open educational resources and related practices within the organisations as well as nationally and internationally are mentioned in the actions.

In the area of research data and methods, the policy, [link](#), brings up that “Finnish operators actively participate in European and international work, e.g. in the European Open Science Cloud (EOSC), the Research Data Alliance (RDA), International Science Council Committee on Data (CODATA) communities, and European research infrastructure work. The Finnish

research community participates actively in dialogue where it can support the openness of research and contribute to the establishment of practical solutions in national and international cooperation.”

Finnish open science monitoring, [link to the monitoring model](#), provides national information required in various international comparisons for open science and research, particularly within Europe. Also, international indicators can be used for monitoring as they evolve and are able to enrich Finnish indicators.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=21>)

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The Finnish Research Infrastructure Committee

The Finnish Research Infrastructure Committee has published the Strategy for National Research Infrastructures in Finland 2020–2030 ([link](#)). It is particularly aimed at promoting the development of national nodes for national and international research infrastructures. Based on the strategy, a roadmap and funding calls are opened by the Committee together with other actors and national and international research infrastructure activities developed in the coming years.

In the strategy, research infrastructures shall make their usage policies easily accessible and provide customer-friendly services for all those interested in their services. The Finnish Research Infrastructure Committee collects information on and disseminates examples of good practices to increase the transparency of research infrastructures.

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.

NO.

-
- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
- xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.